FREE GUIDE · LOAD AS MEDICINE

Joint health on a GLP-1

For adults 50 and over with knee or hip arthritis who take a GLP-1 medicine, or are thinking about one. What the research shows, how to look after your muscle, and what to ask your doctor.

Your doctor decides about medicines

GLP-1 medicines are prescribed by a medical doctor. This guide won't tell you whether to start, stop or change one, and it doesn't compare medicines. It covers how to look after your joints and your muscle, and gives you questions to take to your next appointment.

Stop and contact a clinician if

- your calf hurts or swells after a joint replacement, or you get chest pain, sudden breathlessness, or new trouble controlling your bladder or bowels (call 911 or your local emergency number);
- the joint is hot, red or swollen and you feel feverish or unwell;
- a surgical wound is red, hot, leaking or opening up;
- the joint looks out of shape or cannot take weight after a fall or injury;
- the joint starts giving way, or catching and getting stuck so you can't fully bend or straighten it, or this gets worse;
- you suddenly cannot bend or straighten it as far as usual;
- pain at rest or at night does not ease when you change position, most nights for two weeks or more;
- new numbness, tingling or weakness appears in your leg;
- you lose about 10 pounds or more without trying.

These signs are worth checking before you exercise. They are not a diagnosis.

What a GLP-1 medicine can mean for your joints

You're losing weight on a GLP-1 and your knees still complain on the stairs. So what is the medicine doing for the joint itself?

One large trial asked that question. It enrolled 407 adults with obesity and painful knee arthritis. Everyone got diet and activity counselling. On top of that, each person took either a weekly GLP-1 medicine (semaglutide) or a placebo. After 68 weeks: Bliddal 2024

- **Weight.** People on the medicine lost about 14% of their body weight. People on placebo lost about 3%. ESTABLISHED
- **Pain.** Knee pain scores dropped about 42 points on a 100-point scale with the medicine, and about 28 points with placebo. ESTABLISHED
- **Day-to-day function.** Physical function scores went up 12 points with the medicine and 6.5 points with placebo, also on a 100-point scale. ESTABLISHED
- **Side effects.** About 7 in 100 people stopped the medicine because of side effects, mostly stomach-related. On placebo it was 3 in 100. ESTABLISHED

What this trial can't tell us

- Both groups improved, so the diet and activity support helped too. The medicine group improved more.
- Everyone in the trial was living with obesity, with an average body mass index (BMI) of about 40. The results may not apply to people without obesity.
- It measured pain, function and weight, not cartilage, so it can't tell us whether the joint itself changed. ESTABLISHED
- You may see headlines about these medicines and cartilage. In mice with obesity-related arthritis, semaglutide protected cartilage even when weight loss was taken out of the picture. That is early lab research. It has not been shown in people. UNPROVEN-BRIDGE Qin 2026

Why weight matters to a knee

In a study of older adults with knee arthritis, each pound of weight lost went along with about four pounds less force on the knee with each step while walking. PROMISING Messier 2005

For people with knee arthritis who carry extra weight, losing more than about 5% of body weight over about 20 weeks improved day-to-day function in trials. ESTABLISHED Christensen 2007

Both findings come from weight lost through diet, not medicine. The first is a link seen in one study, not proof of cause and effect.

Protect your muscle

When the scale goes down, not all of what you lose is fat.

On GLP-1 medicines, roughly a quarter to a third of the weight lost is lean mass rather than fat. That is a similar share to weight loss by diet alone. Lean mass includes muscle, but also water and organs, so it isn't all muscle. ESTABLISHED Batsis 2026; Eisa 2026; Jobanputra 2026

Muscle gets you out of a chair, up the stairs and through a day of carrying and lifting, so it is worth protecting while the weight comes off.

Strength training

In weight-loss studies that used diet, people who added strength training kept more of their lean mass. PROMISING Villareal 2017; Eisa 2026 Those studies were not done in people taking a GLP-1, so the benefit there is likely but not yet proven. PROMISING

> **Randy's starting point:** strength training 2 to 3 times a week, working the legs, hips and upper body, and building up gradually. Use the acceptable pain guide later in this guide to judge the dose. CLINICAL GUIDANCE
>
> This is Randy Bauer, PT's clinical guidance, not a research-tested rule for people taking a GLP-1. If you're new to strength training, ask your doctor or a physical therapist how to start.

Protein

Older adults often need more protein than younger adults, especially when losing weight or being active. PROMISING Bauer 2013; Kim 2016 The right amount for you depends on your health, including your kidneys. People with kidney disease may need less, not more.

Ask your doctor or a registered dietitian to set your protein target. If your appetite is smaller on the medicine, ask them how to fit it in.

Exercise is medicine for the joint

International guidelines list arthritis education and structured exercise as core treatments for knee and hip osteoarthritis, recommended for essentially everyone with the condition. ESTABLISHED Bannuru 2019

Whatever you and your doctor decide about medicines, movement stays part of the plan.

The acceptable pain guide

This is the guide Randy uses with his patients. **Rate the ache from 0 to 10.** 0 is no pain at all. 10 is the worst pain you can imagine.

0	1	2	3	4	5	6	7	8	9	10

During activity, many clinicians treat an ache up to about 5 out of 10 as acceptable. By the next day it should be back to your usual. If it is not, do less next time. This is clinical guidance, not a tested safety limit. UNPROVEN-BRIDGE After surgery, follow your surgeon's or physical therapist's protocol. A hot, swollen joint, or one that gets stuck so you can't fully bend or straighten it, needs a clinician, not more exercise. This does not replace advice from a clinician who knows your history.

Talk to your doctor about

Medicines, and any change to them, are prescribed by a medical doctor. Take these points to your next appointment.

- **Gout in the past.** Tell your doctor about any history of gout before starting a GLP-1 medicine, so they can plan with you.
- **A planned surgery or procedure.** If you take a GLP-1 medicine and are planning surgery, including a joint replacement, tell your surgeon and anesthesia team well ahead of time. They will decide with you whether to change your dose or your eating before the operation. ESTABLISHED Kindel 2024
- **Any change to a medicine.** Starting, stopping or changing a medicine, including a GLP-1, is a decision for you and the doctor who prescribes it.
- **Side effects that get in the way** of eating, drinking or staying active.
- **Your exercise plan.** Ask how strength training and the acceptable pain guide fit your plan, and what protein target is right for you.

Notes from my appointment

About the evidence in this guide

Every research statement here carries a grade:

- ESTABLISHED consistent evidence from human studies or guidelines.
- PROMISING good early evidence in people, with limits noted.
- UNPROVEN-BRIDGE a reasoned link, lab research or clinical guidance not yet tested directly in people.
- CLINICAL GUIDANCE Randy Bauer, PT's advice from clinical practice. It is not a research finding.

Sources

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Medical disclaimer

This guide is educational. It does not provide medical advice, diagnosis or treatment, and it is not a substitute for evaluation by a qualified clinician. Talk with your physician or physical therapist before beginning a new exercise program, particularly after a recent surgery. After joint surgery, your surgeon's or physical therapist's protocol comes first. Stop any session for chest pain, dizziness, or unusual shortness of breath, and see the red-flag box at the front of this guide for signs that need a clinician.

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